

STA2020 Tutorial 1: Simple Linear Regression memo

1. Below we have data comparing the actual inventory level to the computer inventory record for different items sold by a company. Both variables are measured in thousands of Rands.

Computer	Actual
2462.35	2339.25
131.25	112.55
232.15	251.45
575.35	575.65
910.05	787.95
1045.45	1029.25
862.55	916.35
1915.55	2009.25
3214.35	3451.05
1209.45	1212.25
245.75	193.15

Table 1: Regression data

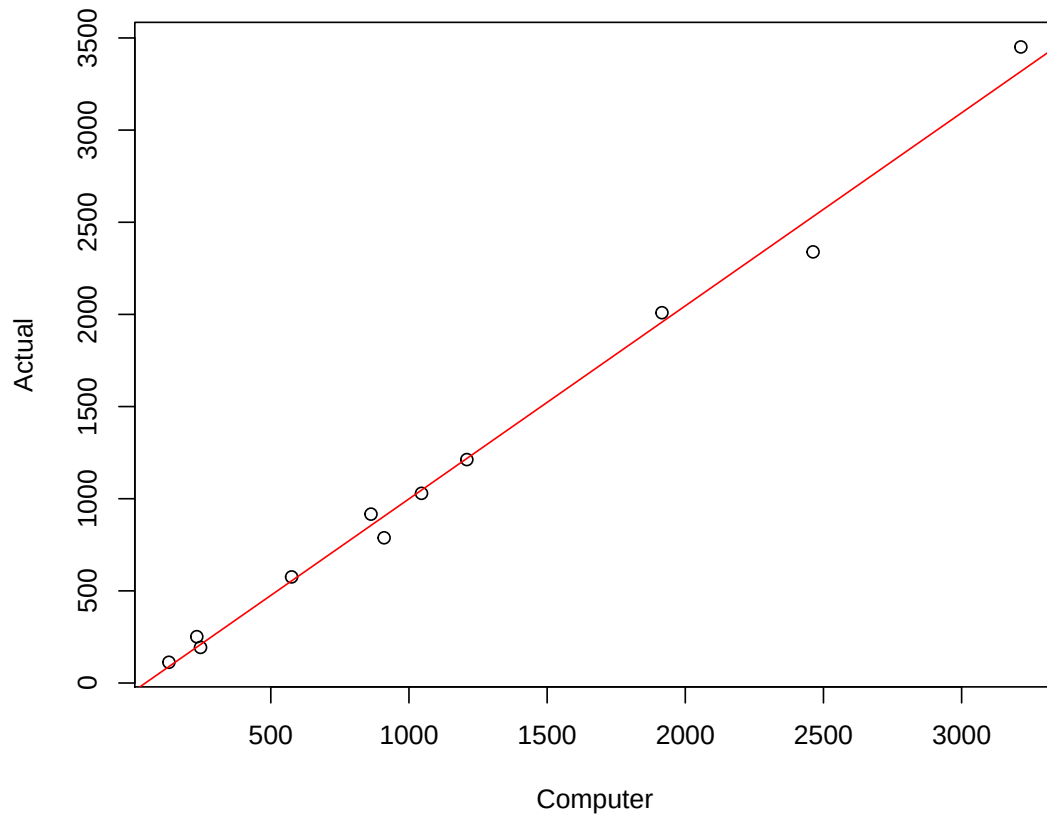
A regression analysis was performed on the data and the following extract of the output obtained:

```
fit <- lm(Actual ~ Computer)
summary(fit)

##
## Call:
## lm(formula = Actual ~ Computer)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -191.36  -16.54   21.49   54.06  132.79
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -48.45898   44.95253  -1.078   0.309
## Computer     1.04740    0.03003  34.883 6.46e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 93.76 on 9 degrees of freedom
## Multiple R-squared:  0.9927, Adjusted R-squared:  0.9918
## F-statistic: 1217 on 1 and 9 DF, p-value: 6.46e-11
```

- (a) Draw a rough scatter diagram for the data provided and comment on the relationship. Do you think running the regression analysis was appropriate?

```
plot(Computer, Actual)
abline(fit, col = "red")
```



Based on the plot there appears to be a strong positive linear relationship between the Computer and Actual values which suggests that a linear regression is appropriate.

- (b) Interpret the coefficient associated with the variable Computer. Is there evidence of a linear association between the variables? In answering this question:
- State the null hypothesis
 - State the alternative hypothesis
 - State the conclusion as well as how you arrived at it.

$\hat{\beta}_{Computer} = 1.05 \implies$ The average change in the Actual value per unit change in the Computer value is 1.05. We test if this coefficient is statistically significant by the hypothesis,

$$H_0 : \beta_{Computer} = 0$$

$$H_1 : \beta_{Computer} \neq 0$$

Based on the test statistic:

$$tval = \frac{\hat{\beta}_{Computer}}{s_{\beta_{Computer}}} = 34.883 \sim t_{n-2}$$

which has an extremely small p-value of 6.46e-11, we reject the null hypothesis and conclude that there is a statistically significant relationship between the Computer and Actual values.

(c) What are residuals and why are they important?

Residuals are the difference between observed and predicted

$$e_i = \hat{y}_i - y_i$$

Residuals are important when determining the quality of a model and the extent to which the data may violate the regression assumptions. You can examine residuals in terms of their magnitude and/or whether they form a pattern.

(d) State the regression assumptions.

- The relationship between x and y is linear,
- The model errors are independent
- The errors $\epsilon \sim N(0, \sigma^2)$ i.e. they are normally distributed with a mean of zero and a constant variance of σ^2 .

2. A study was conducted to determine the association between age and blood pressure (bp). Regression was carried out in R-studio and the resultant output is shown below:

```
agebp <- read.csv("agebp.csv"); attach(agebp)
fit <- lm(bp ~ age)
summary(fit)

##
## Call:
## lm(formula = bp ~ age)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -19.354  -4.797   1.254   4.747  21.153
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  97.0771     5.5276  17.562 2.67e-16 ***
## age          0.9493     0.1161   8.174 8.88e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.563 on 27 degrees of freedom
## Multiple R-squared:  0.7122, Adjusted R-squared:  0.7015
## F-statistic: 66.81 on 1 and 27 DF, p-value: 8.876e-09
```

Use the output shown to answer the following questions. Round all final answers to 2 decimal places.

- (a) What is the estimated regression equation? [1]

$$\hat{b}p = 97.08 + 0.95 * age \checkmark 1$$

- (b) Interpret the meanings of the estimated values of $\hat{\beta}_0$ and $\hat{\beta}_1$? [3]

$\hat{\beta}_0 = 97.08$ i.e. the average blood pressure for an individual who has an age of 0 is 97.08 $\checkmark 1.5$

$\hat{\beta}_1 = 0.95$ i.e. the average increase in blood pressure per unit increase in age/for a 1 year increase in age is 0.95 $\checkmark 1.5$

- (c) Is there evidence of a significant linear relationship between blood pressure and age? Justify your answer by referring to the appropriate values in the output. [3]

Yes there is $\checkmark 1$, the p-value on the coefficient of age tests the null hypothesis that $\beta_1 = 0$, i.e. no relationship. The p-value is extremely small (much smaller than 0.05) $\checkmark 1$ so using an α criteria of 0.05 we reject the null hypothesis and conclude that age does have a significant linear association with blood pressure. $\checkmark 1$

- (d) Use the estimated regression equation to calculate the predicted blood pressure of someone who is 50 years old? [2]

$$\hat{b}p = 97.0771 + 0.9493 * 50 = 144.5421 \approx 144.54 \checkmark \checkmark 2$$

- (e) Calculate a 95% confidence interval for this prediction. [3]

Note that this implies $\alpha = 0.05$ and with $n = 29$, from the t tables we see that $t_{\alpha/2, n-2} = 2.05$. Further it is given that $\bar{x} = 45.97$ and $SSx = 6779.86$. NB The interval can be calculated using

$$y \in \hat{y} \pm t_{\alpha/2, n-2} s_{\varepsilon} \sqrt{\frac{1}{n} + \frac{(x_p - \bar{x})^2}{SSx}}$$

```
144.5421 + 2.05 * 9.563 * sqrt(1/29 + ((50 - 45.97)^2)/6779.86)
## [1] 148.3068

144.5421 - 2.05 * 9.563 * sqrt(1/29 + ((50 - 45.97)^2)/6779.86)
## [1] 140.7774
```